

Julius Zhang

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I am interested in identifying structures and developing new primitives for practical problems.

EXPERIENCE

Columbia University

Researcher, Jun. 2026 - Present

- Conduct research in the Department of Mathematics, hosted by Andrew Blumberg.
- Homotopy theory, arithmetic geometry, cryptography, security, and other topics.

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Researcher, Apr. 2025 - Jul. 2026

- Building [Jolt](#), open-source zero-knowledge virtual machine in Rust.
- Implemented and benchmarked [torus compression](#); proved optimality with a conceptual reframing via Hilbert 90.
- Audited proof systems security: verify correctness of [elliptic-curve optimizations](#) and formalize folklore results into rigorous [proofs](#).
- Evaluated lattice-based cryptography parameters against known attacks under various security regimes and performance constraints.
- Performance engineering across the proving stack, including sum-check optimization.

Millennium

Researcher, Jun. 2024 - Apr. 2025

- Convex optimization and robust statistical estimation for real-time trading systems.

Stellar Development Foundation

Consultant, Jun. 2023 - Sept. 2023

- Byzantine fault tolerance, C++, [stellar-core](#).
- Improved maximum transaction rate per second by 17% via changes to the consensus protocol, reducing number of communication rounds in overlay network. Formal verification in TLA+.

Distributed Systems and TCS Research, Stanford University

Researcher, Sept. 2023 - Jun. 2024

- Under David Mazières, research in general open membership Byzantine fault tolerance protocols ([Github](#)), with applications to certificate authority management and automatic password recovery with hardware security modules.
- Research PCP under Aviad Rubinfeld, reconstructed proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.

Topology Research, Stanford University

Researcher, Jul. 2020 - May 2022

- Floer homotopy theory under Mohammed Abouzaid; related the complex cobordism Gysin map to the Grothendieck residue symbol and derived a recursive formula for Gromov–Witten quantum product corrections.
- Research under Ciprian Manolescu. Floer theory, Seiberg-Witten, knot invariants. Produced new potential counterexamples for the smooth Poincaré conjecture.

SELECTED PAPERS AND MANUSCRIPTS

Julius Zhang. *The Relative Trace-Zero Subgroup of the Barreto-Naehrig Curves*, Cryptology ePrint Archive, 2026. [\[PDF\]](#)

Julius Zhang. *Complex Cobordism and Formal Group Law*, 2024. [\[PDF\]](#)

TALKS

- **p -Ordinary Cohomology for $SL(2)$ over Number Fields**, Hida Theory Seminar, Columbia University, 2026.
- **Construction of Seiberg-Witten-Floer stable homotopy types**, Homotopy Theory and Manifolds Seminar, Columbia University, 2026.
- **Bordered Heegaard Floer Homology and the Surgery Exact Triangle**, Stanford University, 2022.
- **Knots, links, and the search for exotic 4-spheres**, Stanford University, 2021.

EDUCATION

Courant Institute of Mathematical Sciences, New York University, computer science *PhD*. starting Sept. 2026

- Verifiable systems and theoretical computer science.

Stanford University, computer science *M.S.* and mathematics *B.A.*

Sept. 2019 - Jun. 2024

- General Wang Yaowu Scholarship.

SKILLS

Computer Science: Rust, C/C++, Python, formal verification, convex optimization, distributed systems, cryptographic protocol design, performance engineering.

Mathematics: number theory, algebraic geometry, symplectic topology, homotopy theory, Floer theory, gauge theory, concentration inequalities.